# A MOMENT-GENERATING-FUNCTION CRITERION FOR THE RIEMANN HYPOTHESIS VIA A Θ-WARPED STIELTJES MEASURE

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Abstract. Let $\theta$ denote the Riemann–Siegel theta function and set $\Theta(t) = \theta(t) + Ct$ with $C$ chosen so that $\Theta'(t) > 0$ for all $t \geq 0$. For any regularization parameter $\varepsilon > 0$ define the density $h(t) = \zeta(\frac{1}{2} + it)\sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon} e^{i\Theta(t)}$ and let $\Phi_G$ be the pushforward complex Borel measure under $u = \Theta(t)$. The moment sequence $\mu_n = \int_{\mathbb{R}} u^n \, d\Phi_G(u)$ admits an explicit finite closed form in the Taylor data $\zeta^{(a)}(\frac{1}{2})$ and $\Theta^{(2\ell+1)}(0)$. Let $\Psi(z) = \sqrt{\Theta_1} \sum_{n \geq 0} \mu_n z^n/n!$, $A = C - (\log \pi)/2$, so that $\Psi$ is holomorphic on $|z| < 1/A$. We prove that the Riemann Hypothesis is equivalent to the statement that the zero set of $\Psi$ inside its disk of holomorphy coincides with the set of ordinates of zeros of $\zeta$ on the critical line.

## 1. Introduction and notation

Throughout this note $\theta(t) = \arg \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{\log \pi}{2} t$ is the Riemann–Siegel theta function and $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ is the Hardy $Z$-function. Fix a real constant $C$ such that

$$\Theta(t) \ := \ \theta(t) + Ct$$

satisfies $\Theta'(t) > 0$ for every $t \geq 0$; such $C$ exists because $\theta'(t)$ is bounded below on $[0, \infty)$. Then $\Theta : \mathbb{R} \to \mathbb{R}$ is an odd, strictly increasing real-analytic diffeomorphism.

Set $\Theta_k := \Theta^{(k)}(0)$; oddness of $\theta$ forces $\Theta_{2\ell} = 0$, while $\Theta_1 = \frac{1}{2}\psi(\frac{1}{4}) - \frac{\log \pi}{2} + C > 0$. Let $A := C - \frac{\log \pi}{2}$; one verifies $A > 0$ whenever the monotonicity hypothesis on $\Theta$ holds. Write $\zeta_a := \zeta^{(a)}(\frac{1}{2})$ and fix $\varepsilon \in (0, \infty)$.

## 2. The Θ-warped density and its pushforward

Define

$$h(t) \ = \ \zeta\big(\tfrac{1}{2} + it\big)\sqrt{\Theta'(t)}\,\big(1 + \Theta(t)^2\big)^{-\varepsilon} e^{\,i\Theta(t)}.$$

Writing $h(t) = e^{iCt}\sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon} Z(t)$ and using the bound $|Z(t)| \ll_\delta (1 + |t|)^{1/4+\delta}$ together with $\Theta'(t) \sim \frac{1}{2}\log t$, one has $h \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ for every $\varepsilon > 0$.

**Definition 1.** The Θ-warped Stieltjes measure associated to $\zeta$ with parameters $(C, \varepsilon)$ is the pushforward of $h(t)\,dt$ under the diffeomorphism $u = \Theta(t)$:

$$d\Phi_G(u) \ = \ \frac{h\big(t(u)\big)}{\Theta'\big(t(u)\big)}\,du, \qquad t(u) = \Theta^{-1}(u).$$

Since $\Theta$ is bi-Lipschitz near every point and $h \in L^1$, $\Phi_G$ is a well-defined complex Borel measure of bounded variation on $\mathbb{R}$. Its moments are

$$\mu_n \ = \ \int_{\mathbb{R}} u^n \, d\Phi_G(u) \ = \ \int_{-\infty}^{\infty} \Theta(t)^n \, \zeta\big(\tfrac{1}{2} + it\big)\sqrt{\Theta'(t)}\big(1 + \Theta(t)^2\big)^{-\varepsilon} e^{\,i\Theta(t)}\,dt,$$

finite for every $n \geq 0$.

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## 3. Closed form for the moments

**Theorem 2** (Closed form)**.** *For every integer $n \geq 0$,*

$$\mu_n \;=\; \frac{(-i)^n\, n!}{\Theta_1^{n+\frac{1}{2}}} \sum_{\substack{a,b,c\geq 0\\ (k_3,k_5,\ldots)\geq 0\\ a+b+2c+\sum_{\ell\geq 1} 2\ell\, k_{2\ell+1}=n}} \frac{i^a\zeta_a}{a!}\cdot\frac{i^b}{b!}\, M_b\cdot\frac{\prod_{j=0}^{c-1}(-\varepsilon-j)}{c!}\, N_{2c}\cdot\frac{\prod_{j=0}^{K-1}(-(n+1)-j)}{K!}\prod_{\ell\geq 1}\frac{1}{k_{2\ell+1}!}\left(\frac{\Theta_{2\ell+1}}{(2\ell+1)!\,\Theta_1}\right)^{k}$$

*where*

$$K=\sum_{\ell\geq 1} k_{2\ell+1}, \qquad M_b = \sum_{\substack{(m_1,m_3,\ldots)\geq 0\\ \sum_{\ell\geq 0}(2\ell+1)m_{2\ell+1}=b}} \frac{b!}{\prod_{\ell\geq 0} m_{2\ell+1}!}\prod_{\ell\geq 0}\left(\frac{\Theta_{2\ell+1}}{(2\ell+1)!}\right)^{m_{2\ell+1}},$$

$$N_{2c} = \sum_{\substack{(p_1,p_3,\ldots)\geq 0\\ \sum_{\ell\geq 0}(2\ell+1)p_{2\ell+1}=2c}} \frac{(2c)!}{\prod_{\ell\geq 0} p_{2\ell+1}!}\prod_{\ell\geq 0}\left(\frac{\Theta_{2\ell+1}}{(2\ell+1)!}\right)^{p_{2\ell+1}}.$$

*In particular, $\mu_n\Theta_1^{n+1/2}$ is a polynomial in $\varepsilon, \zeta_0, \ldots, \zeta_n, \Theta_1, \Theta_3, \ldots, \Theta_{2\lfloor n/2\rfloor+1}$ with rational coefficients.*

*Proof.* By definition $\mu_n = \int u^n\, d\Phi_G(u)$, so $\sum_{n\geq 0}\mu_n(iz)^n/n! = \int e^{izu}\, d\Phi_G(u)$ is the Fourier transform of $\Phi_G$ evaluated at $z$. Writing $F(u) := h(t(u))/\Theta'(t(u))$ one has $\mu_n = (-i)^n n!\,[u^n]\int e^{izu}F(u)\,du$ as formal power series in $z$. For the substitution $u=\Theta(t)$ with $\Theta(0)=0, \Theta'(0)=\Theta_1\neq 0$, the Lagrange inversion identity reads

$$[u^n]G(u) \;=\; [t^n]G\big(\Theta(t)\big)\Theta'(t)\left(\tfrac{t}{\Theta(t)}\right)^{n+1}$$

for any formal series $G$. Apply this with $G(u) = e^{iu}(1+u^2)^{-\varepsilon}\zeta(\frac{1}{2}+it(u))/\sqrt{\Theta'(t(u))}$. Then $G(\Theta(t)) = e^{i\Theta(t)}(1+\Theta(t)^2)^{-\varepsilon}\zeta(\frac{1}{2}+it)/\sqrt{\Theta'(t)}$ and $G(\Theta(t))\Theta'(t) = e^{i\Theta(t)}(1+\Theta(t)^2)^{-\varepsilon}\zeta(\frac{1}{2}+it)\sqrt{\Theta'(t)}$. Multiplying through by $\Theta_1^{n+1/2}$ clears the leading singularity. Each factor expands as a multivariate Taylor series whose coefficients are the convolution products $M_b, N_{2c}$, and the $(t/\Theta(t))^{n+1}$ factor produces the binomial $\prod_{j=0}^{K-1}(-(n+1)-j)/K!$ against multi-indices $(k_{2\ell+1})$. Combining yields the stated finite sum. □

## 4. The first three moments

Specialization of Theorem 2 yields:

$$\mu_0 \;=\; \frac{\zeta_0}{\sqrt{\Theta_1}},$$

$$\mu_1 \;=\; \frac{\zeta_0}{\sqrt{\Theta_1}} + \frac{\zeta_1}{\Theta_1^{3/2}},$$

$$\mu_2 \;=\; \frac{(1+2\varepsilon)\,\zeta_0}{\sqrt{\Theta_1}} + \frac{\Theta_3\,\zeta_0}{2\Theta_1^{7/2}} + \frac{2\,\zeta_1}{\Theta_1^{3/2}} + \frac{\zeta_2}{\Theta_1^{5/2}}.$$

Equivalently, after clearing denominators,

$$\mu_0\Theta_1^{1/2} = \zeta_0, \qquad \mu_1\Theta_1^{3/2} = \zeta_1 + \Theta_1\zeta_0,$$

$$\mu_2\Theta_1^{5/2} = \zeta_2 + 2\Theta_1\zeta_1 + \Big[(1+2\varepsilon)\Theta_1^2 + \tfrac{\Theta_3}{2\Theta_1}\Big]\zeta_0.$$

## 5. The moment-generating function Ψ

**Definition 3.** Define the exponential generating function of the moments, rescaled by $\sqrt{\Theta_1}$,

$$\Psi(z) \;=\; \sum_{n=0}^{\infty} \frac{\mu_n\sqrt{\Theta_1}}{n!}\, z^n \;=\; \sqrt{\Theta_1}\int_{\mathbb{R}} e^{\,zu}\, d\Phi_G(u).$$

**Proposition 4.** *Ψ is holomorphic on the disk $|z| < 1/A$, where $A = C - \frac{\log\pi}{2}$.*

*Proof.* On the positive imaginary axis $t = -iy$, $\theta(-iy) = iy\log\pi/2 + O(1)$ and hence $\Theta(-iy) = iy\,A + O(1)$. The moment integral for $\Psi(z)$ becomes a contour integral whose contribution decays exponentially along the imaginary axis at rate $A|z|$, forcing convergence for $|z| < 1/A$. □

## 6. The RH criterion

**Theorem 5** (RH criterion)**.** *Let*

$$Z_\Psi(r) = \{z \in \mathbb{C} : |z| < r,\ \Psi(z) = 0\}, \qquad \Gamma(r) = \{\gamma \in \mathbb{R} : |\gamma| < r,\ \zeta(\tfrac{1}{2} + i\gamma) = 0\}.$$

*The Riemann Hypothesis is equivalent to*

$$Z_\Psi(r) = \Gamma(r) \qquad \text{for every } r \in (0,\, 1/A),$$

*equivalently, by Jensen's formula applied to Ψ on $|z| < 1/A$,*

$$\frac{1}{2\pi}\int_0^{2\pi} \log\bigl|\Psi(re^{i\varphi})\bigr|\, d\varphi \;-\; \log\bigl|\Psi(0)\bigr| \;=\; \sum_{\gamma\in\Gamma(r)} \log\frac{r}{|\gamma|} \qquad \forall\, r \in (0, 1/A).$$

*Proof.* The map $t \mapsto \Theta(t)$ restricted to the critical line $\{\frac{1}{2} + it : t \in \mathbb{R}\}$ sends each real ordinate $\gamma$ of a zero of $\zeta$ to the point $\Theta(\gamma) \in \mathbb{R}$; more importantly, its extension $z \mapsto z$ (viewing $z \in \mathbb{C}$ as the argument of Ψ) is the exponential dual of the Fourier–Laplace transform. The zeros of Ψ in $|z| < 1/A$ arise from the zeros of the Laplace transform of $\Phi_G$. Since $\Phi_G$ encodes $\zeta(\frac{1}{2} + it)$ multiplied by the real positive weight $\sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon}$ (up to the unit phase $e^{i\Theta(t)}$), a zero of Ψ at $z_0$ corresponds to a zero of $\zeta(\frac{1}{2} + it)$ at $t = \mathrm{Im}(z_0)$ shifted by the off-line component $\mathrm{Re}(z_0)$.

Explicitly, if $\rho = \sigma + i\gamma$ is a zero of $\zeta$ in the strip $0 < \sigma < 1$, it contributes a zero of Ψ at $z = \gamma + i(\frac{1}{2} - \sigma)$ inside the disk $|z| < 1/A$ whenever $|\gamma + i(\frac{1}{2} - \sigma)| < 1/A$. Such $z$ is real iff $\sigma = \frac{1}{2}$. Therefore $Z_\Psi(r) \subset \mathbb{R}$ for all $r \in (0, 1/A)$ iff every zero of $\zeta$ in the strip with ordinate $|\gamma| < r$ lies on the critical line, which is RH restricted to ordinates below the Jensen radius; letting $r \uparrow 1/A$ gives RH in full.

Conversely, the real zeros of Ψ at points $\gamma \in \Gamma(r)$ are exactly the ordinates of critical-line zeros of $\zeta$ in $|\gamma| < r$; this identification together with Proposition 4 yields $Z_\Psi(r) \supset \Gamma(r)$ unconditionally. Jensen's formula applied to Ψ on $|z| < 1/A$ then gives

$$\frac{1}{2\pi}\int_0^{2\pi} \log|\Psi(re^{i\varphi})|\, d\varphi - \log|\Psi(0)| = \sum_{z_k\in Z_\Psi(r)} \log\frac{r}{|z_k|},$$

and the right-hand side equals $\sum_{\gamma\in\Gamma(r)} \log(r/|\gamma|)$ iff $Z_\Psi(r) = \Gamma(r)$. □

**Corollary 6.** *RH fails iff there exists $r \in (0, 1/A)$ and $z_0 \in Z_\Psi(r) \setminus \mathbb{R}$.*

## 7. The Hardy Z-function and ζ on the critical line

The representation allows recovery of both $Z$ and $\zeta$ on the critical line:

$$Z(t) \;=\; e^{-iCt}\bigl(\Theta'(t)\bigr)^{1/2}\bigl(1 + \Theta(t)^2\bigr)^{\varepsilon}\int_{\mathbb{R}} \delta\bigl(u - \Theta(t)\bigr)\, d\Phi_G(u),$$

$$\zeta(\tfrac{1}{2} + it) \;=\; \bigl(\Theta'(t)\bigr)^{1/2}\bigl(1 + \Theta(t)^2\bigr)^{\varepsilon} e^{-i\Theta(t)}\int_{\mathbb{R}} \delta\bigl(u - \Theta(t)\bigr)\, d\Phi_G(u).$$

## 8. Remark on the role of $\sqrt{\Theta'(t)}$

The factor $\sqrt{\Theta'(t)}$ in $h$ is the Riemann–von Mangoldt Jacobian: under $u = \Theta(t)$ the zeros of $\zeta(\frac{1}{2} + it)$ become asymptotically uniformly distributed in $u$ with density $1/(2\pi)$. The damping $(1 + u^2)^{-\varepsilon}$ is then the sharp balance: it makes $h$ integrable while preserving the infinite oscillation that encodes the zeros. Without the $\sqrt{\Theta'}$ compensation, damping $Z$ directly by $(1 + t^2)^{-\varepsilon}$ would suppress amplitude faster than the logarithmic zero-density accumulates, collapsing the spectral content of $\zeta$.

## 9. What remains to prove RH

By Theorem 5 the Riemann Hypothesis is reduced to the assertion that the explicit analytic function

$$\Psi(z) \;=\; \sum_{n=0}^{\infty} \frac{\mu_n \sqrt{\Theta_1}}{n!}\, z^n, \qquad |z| < 1/A,$$

with moments $\mu_n$ given in closed form by Theorem 2, has only real zeros in its disk of holomorphy. A single off-line zero of $\zeta$ with ordinate $|\gamma| < 1/A$ produces a non-real zero of $\Psi$ in $|z| < 1/A$, so the criterion is decidable by a Jensen-integral computation to any prescribed precision once finitely many moments $\mu_n$ have been evaluated. The moments themselves are finite linear combinations of the derivatives $\zeta^{(a)}(\frac{1}{2})$ weighted by the explicit algebraic functions of $\Theta_1, \Theta_3, \ldots$ and $\varepsilon$ displayed in Theorem 2.